strip. The meter reads the blood sample and displays the glucose level in the units of mg/dL. Most glucometers are based on the electrochemical technology or colour reflectance principle (Figure12.3).

Significance

1. Handy and portable
2. Immediate results, the results are displayed in approximately 40 seconds.
3. Requires no calculation
4. No training is required for operating the instrument

Normal Glucose Values in blood

Normal glucose value:	70 – 100 mg/dL
Random blood sugar:	80 – 120 mg/dL
Fasting blood sugar:	70 – 110 mg/dL
Post – prandial blood sugar:	110 – 140mg/dL

Autoanalyser

Autoanalyser is a computer controlled device. It is used for quick estimation of several bio-chemical parameters like glucose, urea, cholesterol, enzymes and other proteins present in body fluids. The analyser is fully integrated with all the necessary controls such as temperature of reaction, volumes of sample, reagents and transfer of fluids. More sophisticated multichannel autoanalysers with capabilities of selecting any type of analysis for any one set of samples are available now (Figure12.4).



Figure 12.4 Autoanalyser

12.2 Imaging Instruments

EEG – Electroencephalogram

EEG is a test to evaluate the electrical activities of the brain (Figure12.5). Brain cells communicate with each other through electrical impulses. An EEG tracks and records the brain wave patterns. It is a graphical recording of the electrical activities of the cortical and sub cortical regions of the brain. It is recorded by placing the surface electrodes on the scalp region. The electrodes analyse the electrical impulses in the brain and send signals to a computer which records the results. In 1929 Germen scientist Hans Berger was the first to analyse the EEG. Hence, EEG is also known as "Berger Wave". The electrical activity recorded by EEG may have synchronised or desynchronised waves. It has four frequency waves/ rhythms namely alpha, beta, delta and theta waves.

Clinical significance of EEG

1. EEG provides a means to study the functioning of the brain and its coordination with other parts of the body.
2. It is useful in diagnosis of neurological and sleep disorders

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